**Total count:**

The total number of cells in a population can be measured by counting a sample under the microscope. This is called direct microscopic count. This is done by using a specialized counting chamber called Petroff Hausser chamber which is a specially designed slide with a grid. The liquid sample is placed on the grid which has a total area of 1mm$^2$ and divided into 25 large squares. The number of cells in large square is counted and the total number of cells is calculated by multiplying it with a conversion factor based on the volume of the chamber (Figure 6.8).

**Advantages**

This is a quick method of estimating cell numbers.

**Disadvantages**

1. Dead cells are also counted
2. Special microscopes like phase contrast microscope are needed if unstained samples are used.
3. Small cells are difficult to count

## 2. Viable Count

A viable cell is one that is able to divide and form a visible colony on the nutrient media. Viable cells are counted by methods pour plate and spread plate.

**Pour plate method**

In this method, a known volume (0.1 or 1.0ml) of the culture is pipetted into a sterile petri plate, then molten nutrient medium is poured over and incubated. Colonies will appear throughout the agar medium and are counted to obtain viable count.

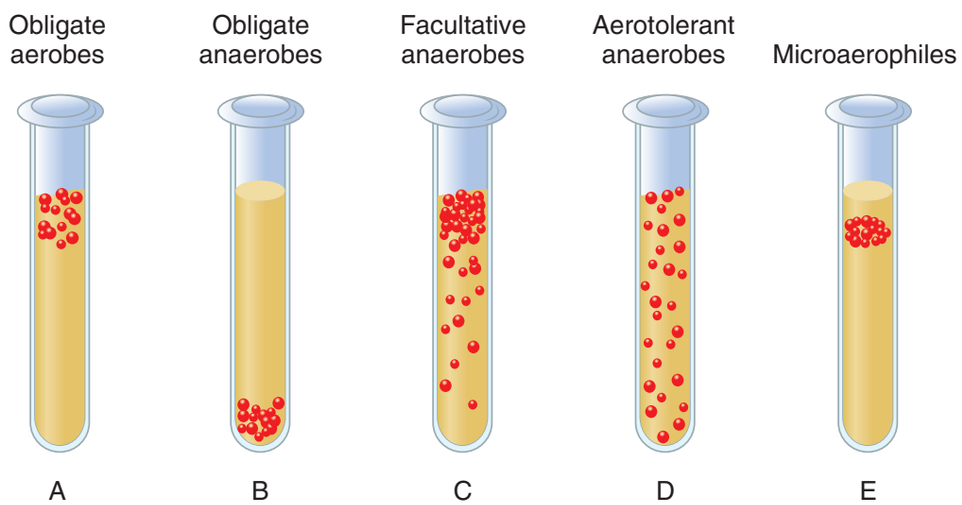



**Figure 6.7:** The effect of oxygen on the growth of various types of bacteria

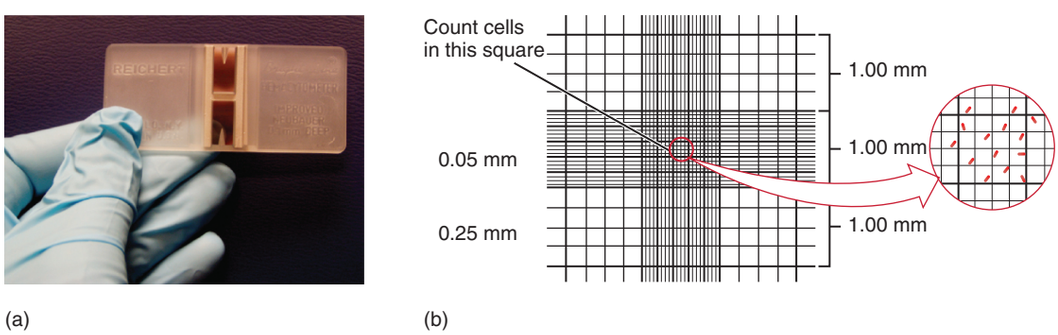



**Figure 6.8:** (a) Petroff-Hausser counting chamber
(b) Microscopic observation of bacterial cells



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